

Modeling and Forecasting VN-Index Volatility: A Robust Empirical Comparison of GARCH-Family, Neural, and Hybrid Approaches

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Course: Time Series

Abstract—This paper evaluates one-step-ahead daily VN-Index volatility forecasts using CafeF VNINDEX data, econometric and neural benchmarks, validation-based model selection, statistical comparison, VaR backtesting, and robustness to OHLC-based volatility proxies. The target is next-day squared percentage log return. On the original 726-observation common test window, GARCH(1,1) is the strongest original benchmark by QLIKE (1.052888), while ARIMA-GARCH is close (1.055679) and statistically indistinguishable by QLIKE Diebold-Mariano testing. A broader GARCH-family search attempts 3,168 specifications and obtains 2,592 successful fits; the validation-selected ARIMA(3,0,3)-EGARCH(1,1,2)-Normal model obtains test QLIKE 1.032548, and the best asymmetric category model obtains 1.029992. Several advanced GARCH-family models improve raw QLIKE relative to GARCH(1,1), but no improvement survives Holm correction at the 10% level, and an approximate model confidence set retains 68 of 73 models. Neural and hybrid forecasts do not robustly dominate econometric benchmarks. Log-target neural variants reduce MAE but severely underpredict volatility spikes and tail risk; calibration improves average log-target QLIKE from 2.225569 to 1.146217, yet remains behind the best GARCH-family forecasts. OHLC proxy analysis shows proxy-sensitive rankings: AdvGARCH-BestAsymmetric has average QLIKE rank 3.29 across seven proxies, compared with 7.00 for GARCH(1,1). Overall, the evidence supports GARCH-family models, especially asymmetric or EGARCH-type specifications, as the most reliable class for daily VN-Index volatility forecasting.

Index Terms—VN-Index, volatility forecasting, GARCH, EGARCH, LSTM, hybrid forecasting, QLIKE, VaR backtesting, range-based volatility.

I. Introduction

Volatility forecasting is risk forecasting, not price forecasting. For an equity index, a one-step variance forecast summarizes the expected magnitude of tomorrow's return shock and therefore supports margin setting, risk limits, Value-at-Risk (VaR), and portfolio allocation. The VN-Index is a natural empirical setting because daily returns exhibit volatility clustering and occasional large moves, while the true latent conditional variance is unobserved.

This study evaluates one-step-ahead daily VN-Index volatility forecasts using daily CafeF VNINDEX observations. The forecasting target is the next-trading-day squared percentage log return. The comparison spans historical and rolling variance baselines, GARCH(1,1), ARIMA-GARCH, LSTM, ARIMA-GARCH-LSTM, tuned neural variants, a broad GARCH-family search, refit

protocols, validation-calibrated neural forecasts, forecast combinations, VaR backtests, and robustness checks based on OHLC range estimators.

The paper is not framed as a contest between deep learning and GARCH. A more useful framing is benchmark reliability under a disciplined empirical protocol. Strong classical baselines matter because volatility targets are noisy, and flexible models can appear attractive under one loss while failing under another. This is especially important when mean absolute error (MAE) rewards smooth low forecasts on calm days but does not penalize volatility-spike underprediction as directly as QLIKE.

The main findings are cautious. GARCH(1,1) is a strong original benchmark on the common test window. Broader validation-based GARCH-family selection and OHLC proxy robustness favor asymmetric or EGARCH-type specifications, but formal statistical superiority over GARCH(1,1) should not be overstated after multiple-testing adjustment. Neural and hybrid forecasts can contain complementary information, especially after calibration, yet they do not robustly dominate GARCH-family forecasts.

The contributions are as follows:

- 1) A reproducible VN-Index volatility forecasting benchmark using daily CafeF data and a one-step-ahead squared-return variance target.
- 2) A broad empirical comparison of simple baselines, GARCH-family models, LSTM, ARIMA-GARCH-LSTM, calibrated neural forecasts, refit protocols, and forecast combinations.
- 3) Statistical forecast comparison using Diebold-Mariano tests, Holm multiple-testing correction, and an approximate model confidence set.
- 4) Risk evaluation through VaR backtesting at 5% and 1% tail levels.
- 5) Robustness analysis using Parkinson, Garman-Klass, Rogers-Satchell, and Yang-Zhang OHLC volatility proxies.
- 6) Evidence that GARCH-family models are the most reliable class in this setting, with advanced asymmetric specifications more robust than the simple GARCH(1,1), while neural and hybrid models do not consistently dominate.

II. Related Work

ARCH and GARCH models provide the classical foundation for conditional volatility forecasting. Engle introduced ARCH to model time-varying conditional variance, and Bollerslev generalized the recursion to include lagged conditional variance [1], [2]. The empirical appeal of GARCH(1,1) is not only parsimony. Large-scale forecast comparisons have shown that it can be difficult to beat in practical volatility forecasting applications [3].

Mean dynamics are often modeled before conditional variance. The Box-Jenkins ARIMA framework captures serial dependence in univariate time series [4]. In financial returns, the conditional mean is usually small relative to the conditional variance, but an ARMA or ARIMA mean model can still be paired with GARCH to test whether residual variance forecasts improve after filtering predictable mean components.

GARCH-family extensions address stylized features that a symmetric GARCH(1,1) recursion may miss. EGARCH models allow asymmetric volatility responses and avoid some positivity restrictions [5]. GJR-type models capture sign-dependent shock effects [6]. APARCH and FIGARCH specifications allow richer power transformations and long-memory volatility dynamics [7], [8]. These extensions motivate the broad specification search evaluated in this paper.

Neural sequence models offer a different forecasting premise. LSTM networks were designed to learn long-range sequential dependencies through gated recurrent states [9]. In volatility forecasting, LSTMs can use lagged returns, volume, rolling volatility, and econometric forecasts as supervised features. Hybrid econometric-neural designs are therefore useful empirical candidates, but their performance must be judged against strong volatility-specific baselines rather than against weak naive alternatives.

Forecast evaluation is central because latent volatility is not directly observed. Squared daily returns are available and simple, but they are noisy proxies for conditional variance. QLIKE is widely used because it remains attractive when forecast comparisons rely on imperfect volatility proxies and because it evaluates forecasts on the variance scale [10]. High-frequency realized volatility can provide richer targets when intraday data are available [11], but daily OHLC estimators are also useful. Parkinson, Garman-Klass, Rogers-Satchell, and Yang-Zhang estimators exploit high, low, open, and close information to obtain alternative volatility proxies [12]–[15].

Statistical forecast comparisons require care when many models are examined. Diebold-Mariano tests compare predictive accuracy under a specified loss difference process [16]. Holm correction controls family-wise error rates across multiple tests [17]. The model confidence set (MCS) framework provides a complementary way to identify a set of models not statistically eliminated under a loss

TABLE I

Chronological data splits and common evaluation window. The common-window comparison aligns all original forecasts on 726 target dates from 2023-02-07 to 2025-12-31.

Dataset	Rows	Origin date range	Target date range
Prepared full	3989	2010-01-04 to 2025-12-31	2010-01-05 to 2025-12-31
Model-ready	3968	2010-02-01 to 2025-12-30	2010-02-02 to 2025-12-31
Train	2471	2010-02-01 to 2019-12-30	2010-02-02 to 2019-12-31
Validation	750	2020-01-02 to 2022-12-29	2020-01-03 to 2022-12-30
Test	745	2023-01-03 to 2025-12-30	2023-01-04 to 2025-12-31
Common test	726	2023-02-06 to 2025-12-30	2023-02-07 to 2025-12-31

criterion [18]. VaR backtesting adds a risk-management view through unconditional and conditional coverage tests [19], [20].

III. Data and Forecasting Target

A. Daily VN-Index Data

The empirical sample uses CafeF VNINDEX daily observations from 2010 through 2025. The cleaned data include open, high, low, close, matched trading volume, and matched trading value. Table I reports the chronological splits. The model-ready sample contains 3968 rows from 2010-02-01 to 2025-12-30 after lagged-return, rolling-volatility, and next-day-target requirements are imposed.

The experimental split is chronological. Training observations end in 2019, validation covers 2020–2022, and the test period begins in 2023. The original common-window comparison uses 726 aligned test observations with target dates from 2023-02-07 to 2025-12-31, so every original model is evaluated on identical forecast origins and labels.

B. Squared-Return Target

Let C_t denote the VN-Index close on trading day t . Percentage log return is

$$r_t = 100 \log \left(\frac{C_t}{C_{t-1}} \right), \quad (1)$$

and the daily squared-return variance proxy is

$$v_t = r_t^2. \quad (2)$$

The supervised one-step target is

$$y_t = v_{t+1} = r_{t+1}^2. \quad (3)$$

Thus a forecast issued at origin date t uses information available through t to predict the target date $t + 1$.

Squared daily return is observable and convenient, but it is a noisy volatility proxy. A single daily return can be small even when intraday variation is high, and a single jump can dominate the proxy. This motivates QLIKE-based evaluation and robustness checks with OHLC range-based volatility proxies.

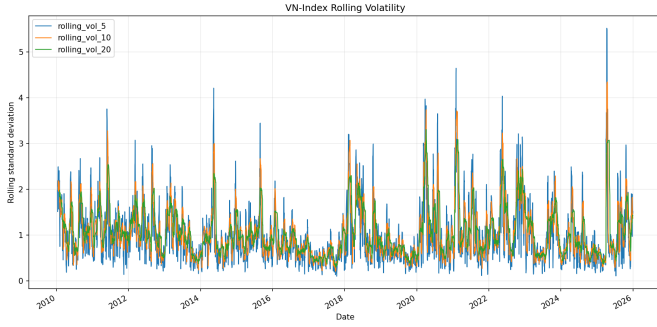


Fig. 1. Rolling VN-Index volatility changes materially across the sample, motivating conditional-variance models rather than constant-variance forecasts.

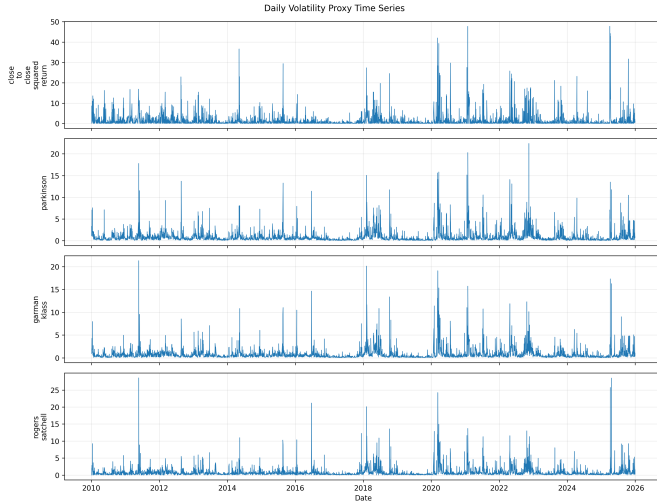


Fig. 2. Alternative volatility proxies have different smoothness and scale, so robustness analysis must distinguish directional conclusions from exact ranking stability.

C. OHLC Volatility Proxies

The proxy robustness analysis constructs seven volatility proxies from the cleaned OHLC sample: close-to-close squared return, Parkinson, Garman-Klass, Rogers-Satchell, and Yang-Zhang rolling proxies over 5, 10, and 20 days. Valid observations range from 3969 for Yang-Zhang 20-day to 3984 for Yang-Zhang 5-day. Correlations with the squared-return proxy are moderate or low: 0.654 for Parkinson, 0.488 for Garman-Klass, 0.345 for Rogers-Satchell, 0.418 for Yang-Zhang 5-day, 0.361 for Yang-Zhang 10-day, and 0.274 for Yang-Zhang 20-day. These values confirm that proxy choice can affect model rankings.

IV. Models

A. Baselines

The historical mean baseline predicts the training-sample average variance. Rolling baselines use 5-, 10-, and 20-day trailing return volatility, squared to produce variance forecasts. These simple baselines are important

because volatility is persistent and recent realized variation can be competitive over short horizons.

B. GARCH(1,1) and ARIMA-GARCH

GARCH(1,1) models conditional variance as

$$r_t = \mu + \epsilon_t, \quad \epsilon_t = \sigma_t z_t, \quad (4)$$

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (5)$$

The parameters are estimated on the training split. Validation and test forecasts use fixed estimated parameters while recursively updating the volatility state with returns observed before each forecast origin.

ARIMA-GARCH first models the conditional mean of returns using ARIMA and then fits GARCH(1,1) to the ARIMA residuals. The implemented ARIMA order selection considers $p, q \in \{0, 1, 2, 3\}$ with $d = 0$ using training AIC, selecting ARIMA(0,0,3) for the original benchmark. This design tests whether filtering mean dynamics adds volatility-forecasting value beyond direct GARCH variance modeling.

C. Advanced GARCH-Family Search

The advanced search evaluates direct ARCH-family and two-step ARIMA-GARCH candidates. It includes ARCH, GARCH, EGARCH, GJR-GARCH, APARCH, and FIGARCH-type specifications with Normal, Student- t , skewed- t , and GED innovation distributions where supported by the fitting library. Candidate selection uses validation QLIKE. Test results are reported only after a model or category has been selected on validation data.

The search attempts 3,168 GARCH-family candidates. It records 2,592 successful fits and 576 failures, all classified as numerical errors such as non-finite forecasts. The best overall validation-selected GARCH-family model is AdvGARCH-BestQLIKE, an ARIMA(3,0,3)-EGARCH(1,1,2)-Normal specification. Category selections also identify the best asymmetric, heavy-tail, and symmetric variants.

D. Neural, Hybrid, Calibration, and Combination Models

The base LSTM uses length-20 sequences of lagged market and volatility features, including returns, squared returns, absolute returns, rolling volatility, volume, and trading value. The ARIMA-GARCH-LSTM hybrid augments these inputs with econometric variance and residual-related features. Tuned neural variants include log-target training and QLIKE-style loss training; selection is by validation QLIKE within the relevant neural group.

Log-target models train on transformed variance targets and invert predictions back to the variance scale. This can stabilize training, but it can also produce low and smooth forecasts if scale is not recovered well. Calibration methods therefore include multiplicative QLIKE scaling, mean matching, nonnegative affine calibration, and isotonic calibration fitted on validation data only. Forecast

combinations include equal-weight means, medians, two-model convex combinations, and stacking weights selected by validation QLIKE.

E. Risk Forecasts

Variance forecasts are mapped to VaR forecasts under a common Normal-quantile rule so that different variance models can be compared under a shared risk backtesting convention. The backtests report violation counts, expected violation counts, Kupiec unconditional coverage statistics, and Christoffersen independence and conditional coverage diagnostics.

V. Experimental Protocol

A. Chronological Evaluation and Leakage Controls

All model development follows the chronological train-validation-test design in Table I. Training data are used for fitting parameters, scalars, and imputers. Validation data are used for neural selection, advanced GARCH-family selection, calibration fitting, refit protocol selection, and combination-weight selection. Test data are reserved for final evaluation.

The target, target date, date, and split labels are excluded from predictive features. Rolling features are trailing-only. LSTM validation and test sequences are built within each split so no sequence crosses a split boundary. Econometric parameters are estimated on training data unless an explicitly documented refit protocol is being evaluated.

B. Metrics and Forecast Comparison

For aligned observations $\{y_t, \hat{y}_t\}_{t=1}^n$, the evaluation uses

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}, \quad (6)$$

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|, \quad (7)$$

$$\text{QLIKE} = \frac{1}{n} \sum_{t=1}^n \left(\log(\hat{y}_t) + \frac{y_t}{\hat{y}_t} \right). \quad (8)$$

Predicted variances are clipped at $\epsilon = 10^{-8}$ before QLIKE computation. QLIKE is central because severe underprediction is penalized through $y_t/\hat{y}_t$. MAE is also reported, but it can be misleading for volatility forecasting when low forecasts perform well on many calm days yet fail during volatility spikes.

Diebold-Mariano tests use the loss-difference convention

$$d_t = L_{m,t} - L_{b,t}, \quad (9)$$

where $L_{m,t}$ is the compared model loss and $L_{b,t}$ is the benchmark loss. A positive mean difference means the compared model is worse than the benchmark. Multiple testing is summarized with Holm correction, and a documented moving-block bootstrap approximation is used for the MCS analysis.

VI. Results

A. Original Benchmark Results

Table II reports the original common-window comparison. GARCH(1,1) has the lowest QLIKE, 1.052888, and the lowest RMSE, 3.648839. ARIMA-GARCH is very close, with QLIKE 1.055679, and the QLIKE Diebold-Mariano test against GARCH gives $p = 0.399560$. This is consistent with limited standalone value from ARIMA mean filtering for next-day variance forecasting.

The LSTM and ARIMA-GARCH-LSTM forecasts do not beat GARCH(1,1) by QLIKE. The log-target variants tell a different metric story: Hybrid-LogTarget-Small has the lowest MAE, 1.108165, but QLIKE rises to 2.232917 and mean predicted variance falls to 0.333670 against mean actual variance 1.211912. This is the clearest example of why MAE alone is not a reliable volatility-forecasting criterion.

B. Econometric Diagnostics

The diagnostic tests support conditional-variance modeling. The ADF test on percentage log returns strongly rejects a unit root with statistic -41.084280 and p-value 0. Ljung-Box tests indicate dependence in returns and squared returns, and the ARCH-LM test on returns gives statistic 480.759753 with p-value 5.69×10^{-97} . The fitted GARCH models converged, and residual ARCH-LM p-values for standardized residuals are high after fitting.

The fitted GARCH(1,1) persistence $\alpha + \beta$ is 0.973264. The ARIMA-GARCH residual-volatility leg has persistence 0.971886. These values indicate strong volatility persistence. The ARIMA-GARCH GARCH leg is fitted to zero-mean residuals, so its mean parameter is not available in the stored residual-GARCH artifact.

C. Advanced GARCH-Family Search

Table III summarizes the selected advanced GARCH-family specifications. The validation-selected overall model, AdvGARCH-BestQLIKE, obtains validation QLIKE 1.611304 and full-window test QLIKE 1.032548. AdvGARCH-BestAsymmetric obtains the best standalone advanced test QLIKE, 1.029992, but that test-best statement is exploratory because the model is validation-selected within the asymmetric category rather than selected as the single overall model by test performance.

Raw QLIKE Diebold-Mariano tests show several advanced GARCH-family improvements against GARCH(1,1). AdvGARCH-BestAsymmetric has raw $p = 0.001534$, and AdvGARCH-BestQLIKE has raw $p = 0.018111$. Table IV shows why the interpretation remains cautious: no QLIKE improvement over GARCH(1,1) survives Holm correction at the 10% level, and the approximate MCS retains 68 of 73 models at both 90% and 95%.

Refit protocols do not materially change the main conclusion. The best selected refit for the best GARCH-family model is fixed train-only. Original GARCH improves only

TABLE II

Original common-window test comparison. GARCH(1,1) is numerically best by QLIKE and RMSE, while log-target neural variants obtain low MAE at the cost of much worse QLIKE and lower mean predicted variance.

Rank	Model	N	RMSE	MAE	QLIKE	Mean predicted variance
1	GARCH(1,1)	726	3.648839	1.393909	1.052888	1.241320
2	ARIMA-GARCH	726	3.671816	1.400256	1.055679	1.236080
3	LSTM	726	3.769716	1.639734	1.161001	1.568589
4	ARIMA-GARCH-LSTM	726	3.763253	1.720113	1.168609	1.701791
5	LSTM-QLIKE	726	3.782485	1.714073	1.185876	1.666980
6	HistoricalMean	726	3.799706	1.449153	1.192259	1.198773
7	Hybrid-QLIKE	726	3.807296	1.862578	1.230508	1.877606
8	RollingVol-20	726	3.901051	1.462196	1.280635	1.208200
9	RollingVol-10	726	3.974129	1.478891	1.309964	1.218541
10	RollingVol-5	726	4.011383	1.492394	1.928009	1.201815
11	Hybrid-LogTarget	726	3.867813	1.109300	2.086360	0.339489
12	Hybrid-LogTarget-Small	726	3.868068	1.108165	2.232917	0.333670
13	LSTM-LogTarget-Small	726	3.888176	1.123546	2.281499	0.342765
14	LSTM-LogTarget	726	3.866882	1.108626	2.301502	0.309520

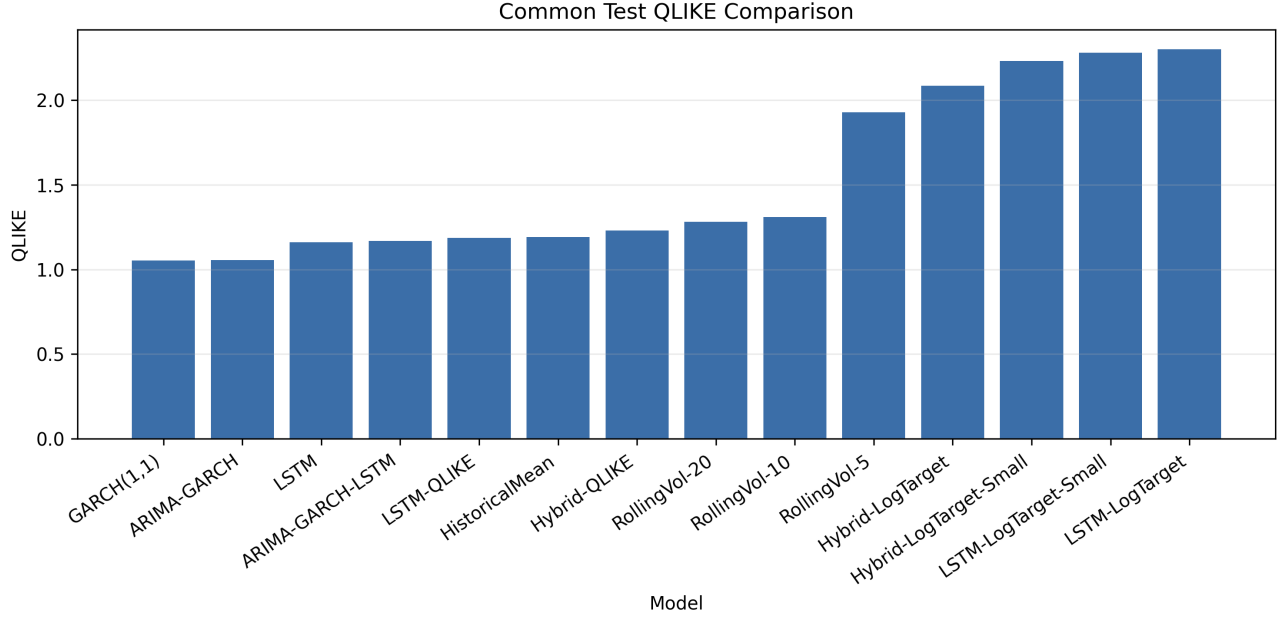


Fig. 3. Common-window QLIKE ranks the original econometric benchmarks ahead of the neural and hybrid variants, while raw log-target models are strongly penalized for variance underprediction.

TABLE III

Validation-selected advanced GARCH-family specifications. Test QLIKE values use the full advanced test window reported in the selected-model tables; statistical tests later align losses on their common window.

Model	Specification	Selection role	Val. QLIKE	Test QLIKE	Paper use
AdvGARCH-BestQLIKE	ARIMA(3,0,3)-EGARCH(1,1,2)-Normal	Overall validation QLIKE	1.611304	1.032548	Primary advanced GARCH-family selected model
AdvGARCH-BestAsymmetric	Zero-mean EGARCH(1,1,1)-Normal	Best asymmetric validation QLIKE	1.612272	1.029992	Category-selected model; best standalone test QLIKE is exploratory
AdvGARCH-BestHeavyTail	ARIMA(3,0,3)-EGARCH(1,1,1)-GED	Best heavy-tail validation QLIKE	1.611777	1.031536	Category-selected heavy-tail comparison
AdvGARCH-BestSymmetric	Constant-mean GARCH(1,3)-Normal	Best symmetric validation QLIKE	1.628556	1.063460	Symmetric robustness comparison

slightly under expanding quarterly refit, with test QLIKE 1.059784 in the refit table, so rolling or expanding refitting is not supported as a robust improvement mechanism here.

D. Neural Calibration and Forecast Combination

Table V reports the calibration and combination evidence. Raw log-target neural forecasts have average test

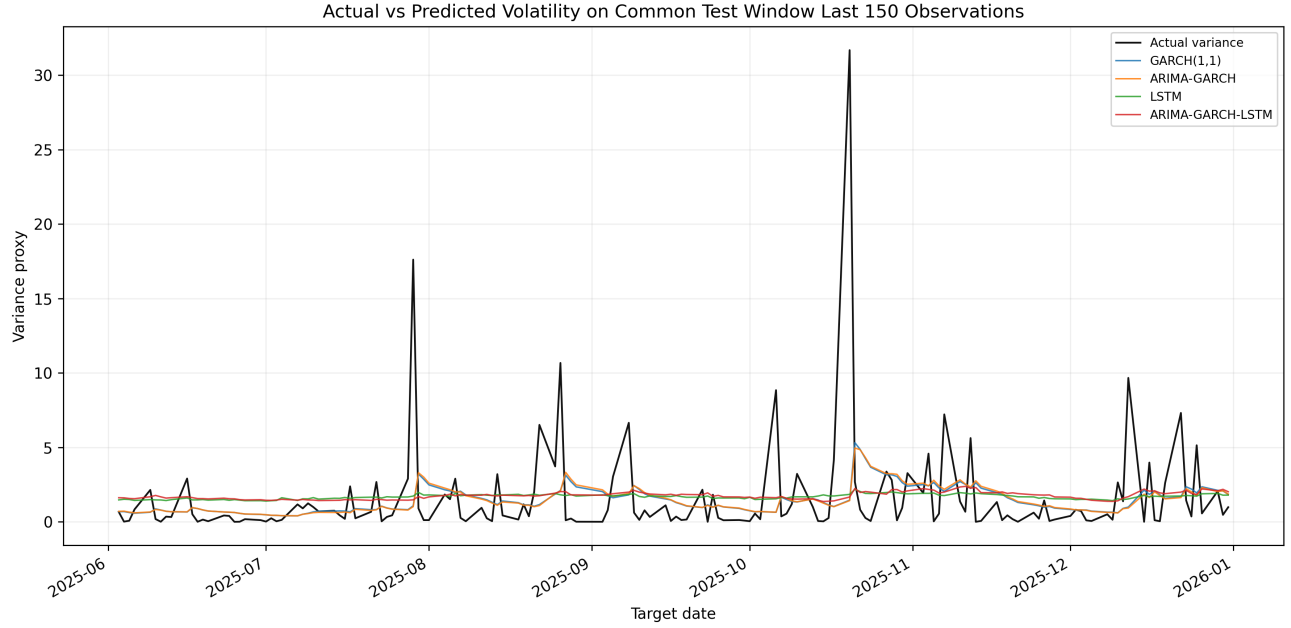


Fig. 4. Actual squared-return variance is spiky, while forecasts are smoother. Low smooth forecasts can reduce MAE on calm days but receive large QLIKE penalties when spikes are missed.

TABLE IV

Statistical comparison summary. Raw QLIKE differences sometimes favor advanced GARCH-family models, but Holm correction and the broad MCS prevent a decisive single-model superiority claim.

Comparison	Evidence	Result	Interpretation
Original QLIKE DM vs. GARCH(1,1)	ARIMA-GARCH $p = 0.399560$; rolling baselines, raw log-target variants, and Hybrid-QLIKE are worse at 10% or stronger	No significant ARIMA-GARCH difference	Mean dynamics add limited standalone value in the original benchmark set
Original squared-error DM vs. GARCH(1,1)	No model differs from GARCH at the 10% level	No robust MSE separation	RMSE rankings should be read descriptively
Original absolute-error DM vs. GARCH(1,1)	Log-target variants beat GARCH by MAE	MAE favors smoother low forecasts	Not sufficient for volatility-risk superiority because QLIKE and risk tests show underprediction
Advanced raw QLIKE DM vs. GARCH(1,1)	AdvGARCH-BestAsymmetric $p = 0.001534$; AdvGARCH-BestQLIKE $p = 0.018111$	Observed improvements before multiplicity correction	Evidence favors advanced GARCH-family forecasts, with caveat
Advanced Holm correction	No QLIKE improvement over GARCH(1,1) survives at 10%	No Holm-significant superiority	Do not claim a statistically decisive winner
Approximate MCS	Moving-block bootstrap approximation retains 68/73 models at 90% and 95%	Broad confidence set	Eliminates weak rolling/log-target models but retains many viable alternatives

TABLE V

Calibration and forecast-combination summary. Calibration corrects much of the log-target scale bias, while validation-selected combinations do not establish robust hybrid superiority.

Result block	Selected or diagnostic item	QLIKE	Pred./actual	Interpretation
Raw log-target neural mean	Raw log-target group	2.225569	0.273	Low MAE comes with severe scale underprediction
Calibrated log-target best	Best validation-calibrated log-target summary	1.146217	1.496	Calibration materially improves QLIKE but remains above best GARCH-family QLIKE
Best validation combination	ComboStack-PoolD_AllStable	1.113530	1.398	Primary combination result; weights mostly calibrated log-target isotonic models
Best observed test combination	BestGARCHFamily + HybridQLIKE pair	1.026265	1.098	Exploratory test-best result; weights 0.73 on AdvGARCH-BestARIMA and 0.27 on Hybrid-QLIKE
Best validation-selected GARCH-family	AdvGARCH-BestQLIKE	1.032548	0.936	Main advanced GARCH-family selected benchmark

TABLE VI

VaR backtesting summary. GARCH-type variance forecasts are closest to nominal at 5%, while raw log-target forecasts substantially understate tail risk.

Model	VaR	N	Viol.	Exp.	Kupiec p
GARCH(1,1)	5%	745	35	37.25	0.7025
ARIMA-GARCH	5%	745	35	37.25	0.7025
LSTM-LogTarget	5%	726	87	36.30	1.54×10^{-13}
ComboStack-	1%	726	10	7.26	0.3337
PoolD_AllStable					
Hybrid-QLIKE	1%	726	10	7.26	0.3337
LSTM-LogTarget	1%	726	56	7.26	3.81×10^{-31}

QLIKE 2.225569 and pred/actual ratio 0.273. Calibration improves the best calibrated log-target QLIKE to 1.146217 and changes the pred/actual ratio to 1.496. This supports scale bias as one cause of poor raw log-target QLIKE, but calibrated neural forecasts still do not beat the best advanced GARCH-family forecasts.

The best validation-selected combination is ComboStack-PoolD_AllStable, with test QLIKE 1.113530. Its stacking weights are concentrated on calibrated log-target isotonic forecasts. The best observed test combination, ComboPair-BestGARCHFamily_HybridQLIKE, has test QLIKE 1.026265 and weights 0.73 on AdvGARCH-BestARIMA and 0.27 on Hybrid-QLIKE. This suggests possible complementary information in hybrid signals, but it is not evidence of robust hybrid superiority because the validation-selected combination trails the best GARCH-family models and the best-test combination is exploratory.

E. VaR Backtesting

Table VI shows that QLIKE performance and VaR coverage are related but not identical. At 5% VaR, GARCH(1,1) and ARIMA-GARCH are closest to nominal, with 35 violations against 37.25 expected and Kupiec $p = 0.7025$. At 1% VaR, Hybrid-QLIKE and the validation-selected combination each have 10 violations against 7.26 expected, with Kupiec $p = 0.3337$.

Raw LSTM-LogTarget is the main risk failure. It has 87 violations at 5% and 56 violations at 1%, far above the expected counts. This aligns with the QLIKE and underprediction evidence: low-MAE log-target forecasts can materially understate risk.

F. Volatility Proxy Robustness

Proxy robustness changes exact rankings. The close-to-close squared-return proxy ranks ComboPair-BestGARCHFamily_HybridQLIKE first by QLIKE. Parkinson, Garman-Klass, Rogers-Satchell, and Yang-Zhang 5-day proxies rank AdvGARCH-BestAsymmetric first. Yang-Zhang 10-day ranks GARCH(1,1) first, tied through ComboPair-GARCH11_ARIMAGARCH, and Yang-Zhang 20-day ranks RollingVol-20 first.

Table VII summarizes rank stability. GARCH(1,1) remains competitive, with average QLIKE rank 7.00 and top-5 placement under 2 of 7 proxies, but it is not consistently top-tier under range-based proxies. AdvGARCH-BestAsymmetric is more stable, with average rank 3.29 and top-3 placement under 5 of 7 proxies. Raw log-target models remain problematic under proxy robustness: their average pred/actual ratio is approximately 0.356 and average spike-underprediction rate is 1.000.

Proxy-specific Holm tests reinforce the cautious interpretation. AdvGARCH-BestAsymmetric has Holm-significant QLIKE improvements over GARCH(1,1) for the close-to-close proxy and for Garman-Klass and Rogers-Satchell; several advanced GARCH variants are Holm-significant under Parkinson. No Yang-Zhang proxy shows a Holm-significant improvement over GARCH(1,1). The stable conclusion is therefore not that one model wins under every proxy, but that advanced asymmetric GARCH-family specifications are the most robust top performers.

VII. Discussion

The results are consistent with the structure of daily equity-index volatility. GARCH-family models directly encode volatility persistence, and the fitted persistence values near 0.97 indicate that this structure is empirically important for the VN-Index. ARIMA mean filtering adds little because the conditional mean of daily returns is small relative to the variance process, so the residual-volatility forecast is close to direct GARCH.

Asymmetric and EGARCH-type variants are favored by the advanced search and by range-based proxy robustness. This is plausible for equity-index returns because negative and positive shocks can have different implications for subsequent risk, and EGARCH specifications can represent asymmetric variance responses without the same positivity constraints as standard GARCH. The evidence is strongest as a robustness pattern rather than as a Holm-corrected declaration of one superior model.

The neural results illustrate a metric trade-off. LSTM and hybrid models are not dismissed: calibrated neural forecasts and pairwise combinations show that nonlinear or hybrid signals can contain information. The issue is reliability under volatility-specific evaluation. Log-target variants can reduce MAE by producing low smooth forecasts, but QLIKE, underprediction analysis, and VaR backtests show that this behavior can miss spikes and understate risk. Calibration reduces scale bias, yet it does not fully solve timing and shape errors.

Volatility proxy choice matters. Squared daily return, Parkinson, Garman-Klass, Rogers-Satchell, and Yang-Zhang proxies do not measure identical objects. The ranking changes under smoother range-based proxies, while class-level conclusions remain broadly stable: GARCH-family forecasts are strongest, asymmetric variants are most robust, and pure neural or hybrid models do not consistently dominate.

TABLE VII

Volatility-proxy robustness rank summary. Range-based proxies favor advanced asymmetric GARCH-family models more often than the simple GARCH(1,1), but exact rankings remain proxy-sensitive.

Model	Avg. QLIKE rank	Best	Worst	Top-5 proxies	Interpretation
AdvGARCH-BestAsymmetric	3.29	1	13	6/7	Most stable top model; top-3 under 5/7 proxies
AdvGARCH-BestHeavyTail	4.43	2	14	5/7	Robust top model, but not a top-1 proxy winner
AdvGARCH-BestQLIKE	4.71	3	10	5/7	Primary validation-selected GARCH-family model remains robust
GARCH(1,1)	7.00	1	15	2/7	Strong benchmark, not consistently top-tier under range proxies
ARIMA-GARCH	9.71	3	19	2/7	Mean dynamics do not improve proxy robustness
ComboPair-BestGARCHFamily_HybridQLIKE	15.14	1	18	1/7	Best close-to-close result, but proxy-sensitive
Hybrid-QLIKE	61.29	58	64	0/7	Pure hybrid QLIKE model is weak under proxy robustness

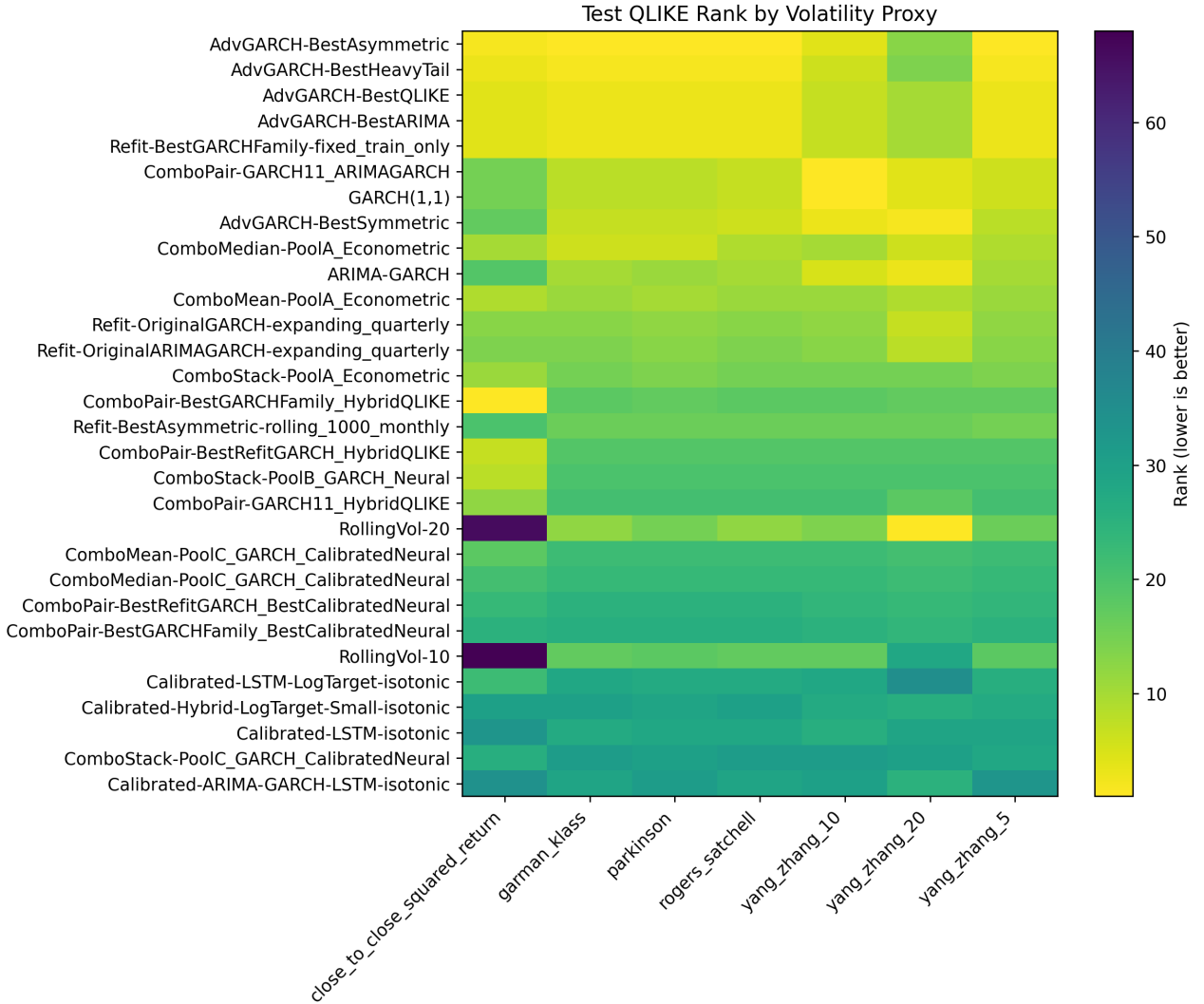


Fig. 5. Proxy-level QLIKE ranks are not invariant. Advanced asymmetric GARCH-family models are more stable across proxies, while several neural and raw log-target variants rank poorly.

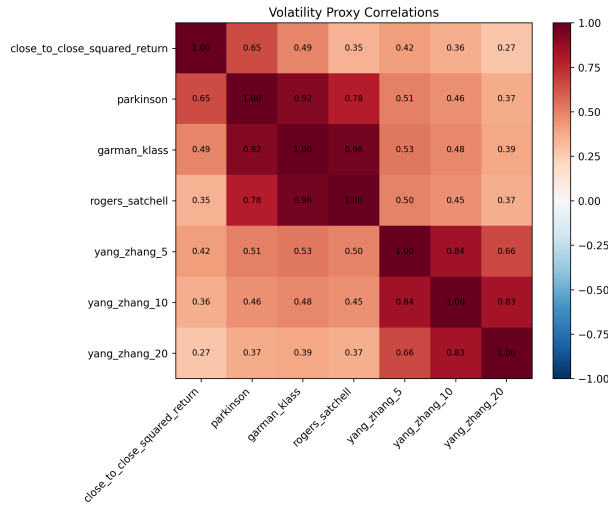


Fig. 6. The moderate and low correlations between squared returns and range-based proxies explain why exact model orderings change across volatility targets.

VIII. Limitations

Several limitations qualify the findings. The study uses daily data only; high-frequency realized volatility is not available. Squared returns and OHLC range estimators are proxies for latent volatility rather than true volatility. The advanced GARCH-family search is broad enough to create validation-overfitting risk, and 576 candidate fits fail numerically. The MCS is a documented moving-block bootstrap approximation and retains many models, so it should be read as conservative evidence against a decisive winner.

The neural models are trained on a modest daily sample for deep learning. Test results cover 2023–2025 and may reflect regime-specific market conditions. Forecast combinations and ex-post best test models are useful diagnostics, but they should not be interpreted as primary selected winners. Finally, all VaR forecasts use a common Normal-quantile mapping, which standardizes comparison but does not exploit distribution-specific tail parameters from fitted GARCH-family models.

IX. Conclusion

This paper provides a robust empirical benchmark for one-step-ahead VN-Index volatility forecasting under validation-based selection, statistical testing, risk back-testing, and volatility-proxy robustness. The main conclusion is that GARCH-family models remain the most reliable class for daily VN-Index volatility forecasting. GARCH(1,1) is a competitive and well-calibrated benchmark, while broader GARCH-family search and OHLC proxy analysis favor asymmetric or EGARCH-type specifications.

ARIMA mean dynamics add limited standalone value. Neural and hybrid models do not robustly dominate econo-

metric benchmarks, although calibrated neural forecasts and selected combinations suggest that hybrid information can be complementary. Log-target neural variants can reduce MAE, but they systematically underpredict volatility spikes and substantially understate tail risk before calibration. The evidence also shows that volatility proxy choice matters: directional conclusions are stable, but exact rankings are not invariant across squared-return and range-based targets.

For applied VN-Index risk forecasting, the practical implication is conservative. Use GARCH-family forecasts as the benchmark class, treat advanced asymmetric specifications as robust candidates, evaluate neural and hybrid forecasts with volatility-specific losses rather than MAE alone, and validate conclusions under alternative volatility proxies and risk backtests before using them for risk management.

References

- [1] R. F. Engle, “Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation,” *Econometrica*, vol. 50, no. 4, pp. 987–1008, 1982. [Online]. Available: <https://www.jstor.org/stable/1912773>
- [2] T. Bollerslev, “Generalized autoregressive conditional heteroskedasticity,” *Journal of Econometrics*, vol. 31, no. 3, pp. 307–327, 1986.
- [3] P. R. Hansen and A. Lunde, “A forecast comparison of volatility models: Does anything beat a garch(1,1)?” *Journal of Applied Econometrics*, vol. 20, no. 7, pp. 873–889, 2005.
- [4] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. John Wiley & Sons, 2015.
- [5] D. B. Nelson, “Conditional heteroskedasticity in asset returns: A new approach,” *Econometrica*, vol. 59, no. 2, pp. 347–370, 1991.
- [6] L. R. Glosten, R. Jagannathan, and D. E. Runkle, “On the relation between the expected value and the volatility of the nominal excess return on stocks,” *The Journal of Finance*, vol. 48, no. 5, pp. 1779–1801, 1993.
- [7] Z. Ding, C. W. J. Granger, and R. F. Engle, “A long memory property of stock market returns and a new model,” *Journal of Empirical Finance*, vol. 1, no. 1, pp. 83–106, 1993.
- [8] R. T. Baillie, T. Bollerslev, and H. O. Mikkelsen, “Fractionally integrated generalized autoregressive conditional heteroskedasticity,” *Journal of Econometrics*, vol. 74, no. 1, pp. 3–30, 1996.
- [9] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [10] A. J. Patton, “Volatility forecast comparison using imperfect volatility proxies,” *Journal of Econometrics*, vol. 160, no. 1, pp. 246–256, 2011.
- [11] T. G. Andersen, T. Bollerslev, F. X. Diebold, and P. Labys, “Modeling and forecasting realized volatility,” *Econometrica*, vol. 71, no. 2, pp. 579–625, 2003.
- [12] M. Parkinson, “The extreme value method for estimating the variance of the rate of return,” *The Journal of Business*, vol. 53, no. 1, pp. 61–65, 1980.
- [13] M. B. Garman and M. J. Klass, “On the estimation of security price volatilities from historical data,” *The Journal of Business*, vol. 53, no. 1, pp. 67–78, 1980.
- [14] L. C. G. Rogers and S. E. Satchell, “Estimating variance from high, low and closing prices,” *The Annals of Applied Probability*, vol. 1, no. 4, pp. 504–512, 1991.
- [15] D. Yang and Q. Zhang, “Drift-independent volatility estimation based on high, low, open, and close prices,” *The Journal of Business*, vol. 73, no. 3, pp. 477–491, 2000.
- [16] F. X. Diebold and R. S. Mariano, “Comparing predictive accuracy,” *Journal of Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995.

- [17] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979. [Online]. Available: <https://www.jstor.org/stable/4615733>
- [18] P. R. Hansen, A. Lunde, and J. M. Nason, "The model confidence set," *Econometrica*, vol. 79, no. 2, pp. 453–497, 2011.
- [19] P. H. Kupiec, "Techniques for verifying the accuracy of risk measurement models," *The Journal of Derivatives*, vol. 3, no. 2, pp. 73–84, 1995.
- [20] P. F. Christoffersen, "Evaluating interval forecasts," *International Economic Review*, vol. 39, no. 4, pp. 841–862, 1998.